

## *Kubernetes Basics Overview of Kubernetes*

1. Install Minikube on your local machine and start a new Kubernetes cluster using the minikube start command. Take a screenshot of your running cluster status.

### ANS

- I installed Minikube on my local machine and started a Kubernetes cluster using the following commands:

- minikube start

- To check the running cluster status:

- minikube status

- To verify that Kubernetes is working:

- kubectl get nodes

- Expected output:

| NAME     | STATUS | ROLES         | AGE | VERSION |
|----------|--------|---------------|-----|---------|
| minikube | Ready  | control-plane | ... | ...     |

2. Write a YAML configuration file named music-pod.yaml to deploy a pod running the nginx image with the pod name set to spotify-nginx. Apply this configuration using kubectl.

### ANS

- I created a YAML configuration file named music-pod.yaml.

- apiVersion: v1
- kind: Pod
- metadata:
- name: spotify-nginx
- spec:
- containers:
  - name: nginx
- image: nginx
- ports:
  - containerPort: 80

- Save the file as:

- music-pod.yaml

- Apply the configuration using:

- kubectl apply -f music-pod.yaml

- Verify the pod:

- kubectl get pods

- Expected output:

| NAME          | READY | STATUS  | RESTARTS | AGE |
|---------------|-------|---------|----------|-----|
| spotify-nginx | 1/1   | Running | 0        | ... |

3. Use kubectl to check the status of your spotify-nginx pod and display its details. Then, delete the pod using the appropriate kubectl command.  
**Hint:** Use kubectl get pods and kubectl describe pod, then kubectl delete pod.

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- To check the status of the spotify-nginx pod:
    - kubectl get pods
  - To display detailed information about the pod:
    - kubectl describe pod spotify-nginx
  - The describe command displays information such as the pod IP, container image, ports, events, node, and container status.
  - To delete the pod:
    - kubectl delete pod spotify-nginx
  - Verify that the pod has been deleted:
    - kubectl get pods
4. Create a YAML file named insta-service.yaml to expose your spotify-nginx pod using a Kubernetes Service of type NodePort, making it accessible externally. Apply the service and note the port assigned.  
**Hint:** Use kind: Service and type: NodePort in your YAML.

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- I created a YAML file named insta-service.yaml to expose the spotify-nginx pod using a Kubernetes Service of type NodePort.
  - apiVersion: v1
  - kind: Service
  - metadata:
  - name: spotify-nginx-service
  - spec:
  - type: NodePort
  - selector:
  - app: spotify-nginx
  - ports:
    - protocol: TCP
  - port: 80
  - targetPort: 80
- **Important:** The Service selector must match a label on the Pod. Therefore, the Pod YAML should contain the following label:
- To get the exact URL in Minikube:
  - minikube service spotify-nginx-service --url
- Open the displayed URL in a browser to access the Nginx web page.

5. Compare deploying a web app using Kubernetes pods and services versus running it directly on a single VM (like a traditional deployment). List 3 advantages of using Kubernetes for scaling an app like Zomato or Flipkart.

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#### **Kubernetes Pods & Services**

Applications run inside containers/pods and can be replicated easily.

Services provide stable networking and load distribution between pods.

Kubernetes can automatically scale applications by increasing the number of pods.

#### **Single VM Deployment**

Application normally runs directly on one VM.

Networking and traffic management must be configured manually.

Scaling usually requires manually creating or resizing VMs.

#### **❖ Advantages of Kubernetes for Applications like Zomato or Flipkart**

1. **Scalability:** Kubernetes can run multiple replicas of an application and increase or decrease pods according to demand.
2. **High Availability:** If one pod fails, Kubernetes can restart it or route traffic to other healthy pods.
3. **Load Distribution:** Kubernetes Services provide a stable endpoint and distribute traffic among available application pods.